

RAG PDF Loader Practice Document

A clean text-based PDF created specifically for practicing LangChain PDF document loading.

1. What is RAG?

Retrieval-Augmented Generation (RAG) is a technique that combines information retrieval with a language model. A RAG application retrieves relevant information from external documents and provides that information to an LLM as context before generating an answer.

2. Basic RAG Pipeline

A typical RAG pipeline follows these steps:

- Load documents from sources such as TXT, PDF, DOCX, or web pages.
- Split large documents into smaller chunks.
- Create embeddings for the chunks.
- Store embeddings in a vector database.
- Retrieve relevant chunks for a user question.
- Send the retrieved context to a language model.
- Generate a grounded answer.

3. PDF Document Loading

LangChain provides document loaders for different file types. A PDF loader extracts text from PDF pages and represents the extracted content as Document objects.

Example using PyPDFLoader

```
from langchain_community.document_loaders import PyPDFLoader

loader = PyPDFLoader("rag_pdf_loader_practice.pdf")
documents = loader.load()

print("Number of pages:", len(documents))
print(documents[0].page_content)
```

4. Understanding the Document Object

Each loaded page is represented as a Document object. The `page_content` field contains the extracted text, while metadata can contain information such as the source file and page number.

Example:

```
document = documents[0]
print(document.page_content)
print(document.metadata)
```

5. Important RAG Terms

Term	Meaning
Document	Original source of information.
Chunk	Smaller section created from a document.
Embedding	Numerical representation of text meaning.
Vector Database	Database used to store and search embeddings.
Retriever	Component that finds relevant chunks.
LLM	Language model that generates the final response.
Context	Retrieved information supplied to the LLM.

6. Practice Questions

1. How many pages does this PDF contain?
2. What does PyPDFLoader return after calling load()?
3. What is stored in page_content?
4. What is the purpose of metadata?
5. Why are documents split into chunks in a RAG system?
6. What is the role of a retriever?
7. What is the difference between a document and a chunk?

7. Next Learning Step

After successfully loading this PDF, the next practical step is to learn text splitting with RecursiveCharacterTextSplitter. After splitting, you can move on to embeddings, vector databases, retrieval, and finally a complete RAG chain.